

Hardware Business Management System

Purpose of This Document

Functional requirements, non functional requirements, validation, error handling, offline behavior, synchronization, reports, security, and acceptance expectations. This standalone file is written for handoff to a Cloud AI, software developer, designer, database planner, or project reviewer.

How to Use This Document

- Use Confirmed requirements as agreed scope for planning and implementation.
- Use Proposed requirements as recommended starting points that can still be adjusted.
- Use Pending Decision items as questions for the business owner before coding or final schema design.
- Do not treat unresolved accounting, profit, allocation, stock costing, tax, or debt rules as final until the owner confirms them.

Status Label Convention

Confirmed means the item is part of the agreed scope. Proposed means it is a recommended implementation detail that can still change. Pending Decision means the business owner must decide before a developer or Cloud AI treats it as final.

Status	Meaning	Developer Handling
Confirmed	Part of the agreed project scope.	Implement unless later changed by the owner.
Proposed	Recommended approach or useful feature.	Validate during design and planning.
Pending Decision	Business rule or accounting detail not fully decided.	Ask the owner or make configurable. Do not hard code.

Document 03 Software Requirements Specification

This SRS translates the business requirements into software requirements for design and development. Each requirement includes an identifier, platform target, status, and acceptance expectation so a developer or Cloud AI can generate tasks, screens, APIs, database schema, and tests from it.

Product Perspective

The system consists of a mobile application, a web administration interface, a backend API, a central MySQL database, a local SQLite database on mobile, and a synchronization layer. The mobile application is expected to work in the shop during normal operations. The web admin is expected to provide central control, reports, configuration, and oversight.

User Classes

User Class	Description	Primary Platform	Status
Super Admin	Business owner or central administrator with full control over shops, users, configuration, reports, and audit trail.	Web and mobile where appropriate	Confirmed
Shop Admin	Shop-level user who records transactions and performs daily opening and closing for assigned shop.	Mobile first, web where appropriate	Confirmed
Future Accountant	Optional user who may review financial records without full system control.	Web	Proposed

Functional Requirements

ID	Module	Requirement	Platform	Status	Acceptance
FR-001	Authentication	Users must log in securely before accessing system data.	Mobile Web	Confirmed	Invalid credentials are rejected and successful login opens only authorized data.
FR-002	Password reset	The system should support secure password reset or admin-initiated password reset.	Web Mobile	Proposed	A user can regain access without exposing passwords.
FR-003	Role permissions	The system must enforce Super Admin and Shop Admin permissions.	Web Mobile	Confirmed	Shop Admin cannot access unauthorized shop or admin-only configuration.
FR-004	Shop management	Super Admin must create, edit, deactivate, and view shops.	Web	Confirmed	A shop can be managed without deleting historical records.
FR-005	User management	Super Admin must create users, assign roles, assign shops, deactivate users, and review user status.	Web	Confirmed	Assigned users can access only the correct shop and features.
FR-006	Product categories	The system must manage product categories for hardware products.	Web Mobile	Confirmed	Products can be grouped and filtered by category.
FR-007	Product records	The system must manage product name, SKU or code, unit, category, buying price, selling price, reorder level, and active status.	Web Mobile	Confirmed	Product details can be added, edited, searched, and deactivated.
FR-008	Purchase recording	The system must record purchases with product, quantity, unit cost, supplier if used, shop, date, payment status, and notes.	Mobile Web	Confirmed	Saved purchase increases stock and appears in purchase reports.
FR-009	Supplier records	The system should maintain supplier contacts where purchases and supplier debts are used.	Web Mobile	Proposed	Supplier can be linked to purchases and filtered in reports.
FR-010	Sales recording	The system must record sales with product, quantity, unit price, discount if enabled, payment method, customer if needed, date, and user.	Mobile Web	Confirmed	Saved sale reduces stock and appears in sales reports.
FR-011	Receipt support	The system should generate a printable or shareable sales receipt.	Mobile Web	Proposed	Receipt includes shop, date, items, totals, payment details, and sale reference.
FR-012	Stock balance	The system must show current stock by product and shop.	Mobile Web	Confirmed	Stock balance changes after purchases, sales, and adjustments.
FR-013	Stock adjustment	The system must allow authorized stock adjustments with reason and audit trail.	Web Mobile	Confirmed	Adjustment records before quantity, after quantity, user, time, and reason.
FR-014	Low stock alert	The system must notify authorized users when product stock reaches reorder level.	Mobile Web	Confirmed	Low stock products appear in alerts and reports.
FR-015	Expense recording	The system must record expense category, amount, date, shop, payment method if used, user, and reason.	Mobile Web	Confirmed	Expenses appear in daily closing and expense reports.
FR-016	Expense categories	Super Admin must manage expense categories.	Web	Confirmed	Categories can be added, edited, deactivated, and used in expenses.
FR-017	Capital entries	The system must record capital injections, withdrawals, and capital balance movement.	Web Mobile	Confirmed	Capital report shows opening, movement, and current capital balance.
FR-018	Debt records	The system must record debts, repayments, outstanding balances, party name, shop, date, and notes.	Mobile Web	Confirmed	Repayment reduces outstanding balance and debt history remains visible.
FR-019	Customer records	The system should maintain customer records for credit sales and debt follow-up.	Mobile Web	Proposed	Customer can be linked to sales, debts, and repayments.
FR-020	Profit calculation	The system must calculate profit once the owner approves the formula and cost method.	Web Mobile	Pending Decision	Profit output matches the approved formula in test cases.

ID	Module	Requirement	Platform	Status	Acceptance
FR-021	60 40 allocation	The system must support a 60 percent and 40 percent allocation after profit is finalized.	Web Mobile	Confirmed	Allocation can be generated for a selected period, with exact accounts pending decision.
FR-022	Daily opening	A Shop Admin must be able to open a business day for a shop.	Mobile Web	Confirmed	Transactions are associated with the active business day.
FR-023	Daily closing	A Shop Admin must be able to close a business day with totals, notes, and exceptions.	Mobile Web	Confirmed	Closing report stores totals and prevents silent editing without audit trail.
FR-024	Offline storage	The mobile app must store transactions locally in SQLite when offline.	Mobile	Confirmed	A transaction can be saved without internet and marked for sync.
FR-025	Synchronization	The mobile app must synchronize local SQLite data with central MySQL when online.	Mobile Backend	Confirmed	Pending records sync successfully and sync status is visible.
FR-026	Conflict handling	The system must detect sync conflicts and resolve them through approved rules.	Backend Mobile Web	Pending Decision	Conflict cases are logged and not silently overwritten.
FR-027	Notifications	The system must notify authorized users about low stock, sync failures, daily closing reminders, and important debt events.	Mobile Web	Confirmed	Relevant notifications appear to the correct users.
FR-028	Reports	The system must provide sales, purchases, stock, expenses, debts, capital, profit, daily closing, and audit reports.	Web Mobile	Confirmed	Reports can be filtered by date, shop, user, product, and category where relevant.
FR-029	Export	The system should export selected reports to PDF or spreadsheet format.	Web	Proposed	A filtered report can be exported for business review.
FR-030	Audit trail	The system must log critical create, update, delete, login, sync, approval, and closing actions.	Backend Web	Confirmed	Audit log shows actor, action, time, before or after values where relevant, and device/source.
FR-031	Settings	Super Admin must configure business name, shops, categories, roles, permissions, stock thresholds, and financial rule settings.	Web	Confirmed	Settings changes are saved and audited.
FR-032	Search and filters	The system must support search and filtering for products, sales, purchases, debts, reports, and audit logs.	Web Mobile	Confirmed	Users can find records quickly by common criteria.
FR-033	Data backup	The system should support backup and restore procedures for central data.	Backend	Proposed	Backup process is documented and tested before production.
FR-034	Record correction	The system must support corrections without destroying audit history.	Web Mobile	Confirmed	Corrections are linked to the original record or logged in audit trail.

Non Functional Requirements

ID	Category	Requirement	Status
NFR-001	Availability	Mobile shop operations must continue offline for core transaction recording.	Confirmed
NFR-002	Performance	Common mobile actions such as product search and sale entry should respond quickly under normal shop data volume.	Proposed
NFR-003	Security	Passwords must be hashed, sessions must expire, and access must be role based.	Confirmed
NFR-004	Privacy	Sensitive user and business data must be visible only to authorized users.	Confirmed
NFR-005	Reliability	Synchronization must retry failed records and preserve a visible sync status.	Confirmed
NFR-006	Integrity	Stock, sales, purchases, debts, expenses, and capital records must preserve referential integrity.	Confirmed
NFR-007	Usability	Shop screens must be optimized for fast entry with minimal typing where possible.	Confirmed
NFR-008	Maintainability	The codebase should separate mobile, web, backend, sync, and reporting responsibilities.	Proposed
NFR-009	Scalability	The design should support multiple shops and growing product and transaction volume.	Confirmed
NFR-010	Auditability	Critical changes must remain traceable after correction or synchronization.	Confirmed

Validation Requirements

- Required fields must be validated before saving records.

- Quantities, prices, costs, expenses, repayments, and capital values must be numeric and non-negative unless a correction workflow allows otherwise.
- Selling below cost may be blocked or warned depending on owner decision.
- A sale should validate available stock unless negative stock is enabled.
- Dates must be valid and assigned to the correct business day.
- Duplicate product codes should be blocked or resolved before saving.
- Offline records must have stable local identifiers so they can synchronize safely later.

Error Handling Requirements

- Network failures must not delete or corrupt local mobile data.
- Failed sync records must remain visible with retry status.
- Users should receive clear messages for validation errors, permission errors, and sync failures.
- Server errors must be logged for administrators without exposing sensitive technical details to Shop Admin users.
- Duplicate sync submissions must be detected to avoid double counting sales, purchases, expenses, repayments, or stock adjustments.